HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Designed and implemented low-level system components for a secure, multi-tenant virtualization platform, contributing to the core functionality of container and function-based services.
- Developed and maintained **Python** scripts for automating operational tasks and data ingestion, improving efficiency in routine system maintenance.
- Contributed to the development of **Java Spring Boot** REST APIs for internal tooling and system integrations, focusing on maintainability and clear contracts.
- Wrote and optimized **SQL** queries against large datasets for data validation and system configuration, ensuring data integrity across enterprise platforms.
- Managed and automated operational aspects of ephemeral compute platforms, including system monitoring and incident response, to ensure service reliability.
- Contributed to the **Firecracker** open-source project by implementing new features and addressing reported issues, collaborating with a global community of developers.
- Containerized microservices using **Docker** and deployed them to **AWS EC2** instances, managing configurations and ensuring consistent environments.

SKILLS

Languages – Java, Python, SQL, C, C++, Rust

Cloud Platforms – AWS (EC2, S3, Lambda, CloudWatch)

Virtualization & Containerization – Firecracker, Docker, Kubernetes

Operating Systems – Linux

Databases – PostgreSQL, MySQL, Oracle

Developer Tools – Git, Jenkins, Jira, Confluence

PROJECTS

Secure Ephemeral Compute Platform Development

- Engineered core virtualization components in **C++** for an open-source microVM technology, focusing on secure isolation for multi-tenant workloads.
- Implemented device model abstractions and guest interaction handlers, enabling efficient resource management and communication channels.
- Integrated new device drivers and I/O path optimizations, contributing to a **15%** reduction in observed guest boot times.

Automated System Health Monitoring

- Developed **Python** scripts to parse logs from distributed systems using **Splunk**, identifying and alerting on critical errors impacting service availability.
- Built automated data pipelines to aggregate operational metrics from multiple services into a centralized reporting database.
- Created scheduled reports summarizing system health trends, providing actionable insights to operations teams for proactive issue resolution.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)